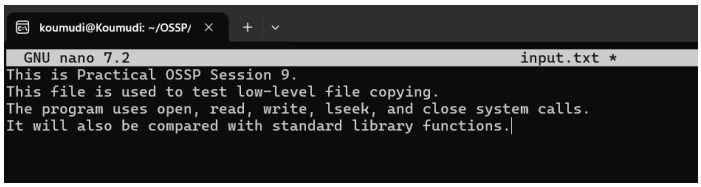
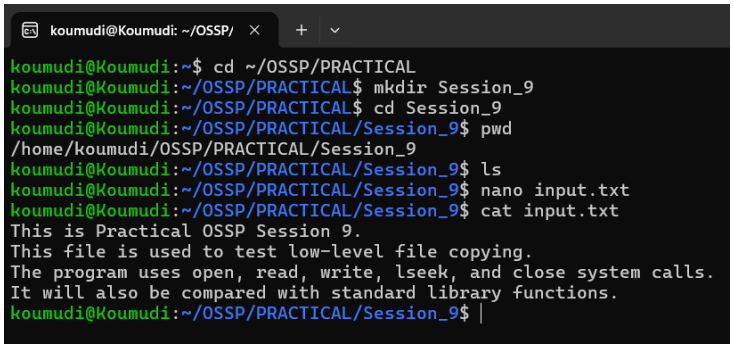
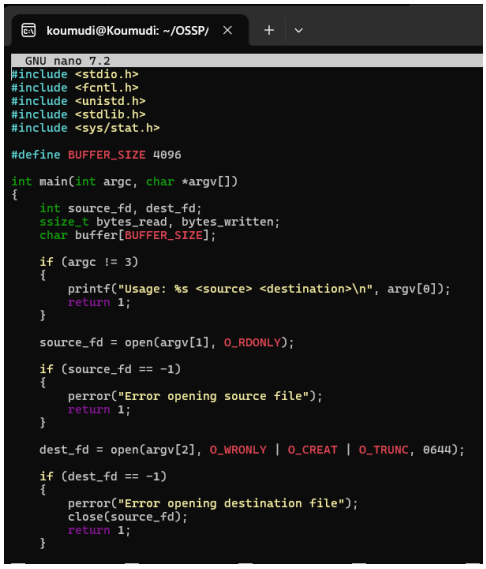
**Practical OSSP**

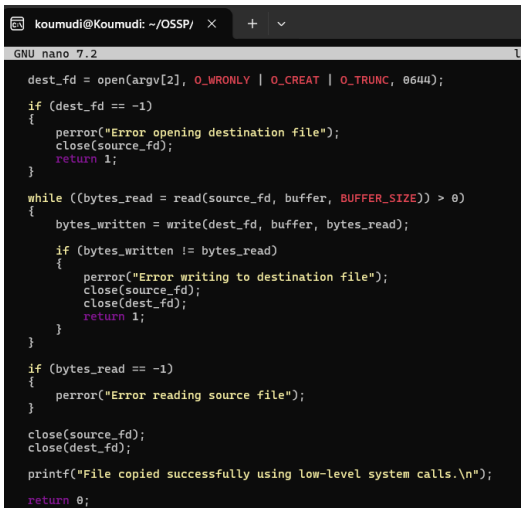
**SESSION 9**

**TASK1:** Implement a file copy utility using low-level Linux file 1/0 system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).



```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ nano low_level_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls
input.txt  low_level_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ./low_level_copy input.txt output_low.txt
-bash: ./low_level_copy: No such file or directory
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ gcc low_level_copy.c -o low_level_copy
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls
input.txt  low_level_copy  low_level_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ./low_level_copy input.txt output_low.txt
File copied successfully using low-level system calls.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cat output_low.txt
This is Practical OSSP Session 9.
This file is used to test low-level file copying.
The program uses open, read, write, lseek, and close system calls.
It will also be compared with standard library functions.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp input.txt output_low.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$
```

koumudi@Koumudi: ~/OSSP/

```
 GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>

#define BUFFER_SIZE 4096

int main(int argc, char *argv[])
{
    FILE *source, *destination;
    char buffer[BUFFER_SIZE];
    size_t bytes_read, bytes_written;

    if (argc != 3)
    {
        printf("Usage: %s <source> <destination>\n", argv[0]);
        return 1;
    }

    source = fopen(argv[1], "rb");

    if (source == NULL)
    {
        perror("Error opening source file");
        return 1;
    }

    destination = fopen(argv[2], "wb");

    if (destination == NULL)
    {
        perror("Error opening destination file");
        fclose(source);
        return 1;
```

koumudi@Koumudi: ~/OSSP/

```
 GNU nano 7.2                                                standard_c
    }

    destination = fopen(argv[2], "wb");

    if (destination == NULL)
    {
        perror("Error opening destination file");
        fclose(source);
        return 1;
    }

    while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, source)) > 0)
    {
        bytes_written = fwrite(buffer, 1, bytes_read, destination);

        if (bytes_written != bytes_read)
        {
            perror("Error writing to destination file");
            fclose(source);
            fclose(destination);
            return 1;
        }
    }

    fclose(source);
    fclose(destination);

    printf("File copied successfully using standard library functions.\n");

    return 0;
}
```

```
It will also be compared with standard library functions.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp input.txt output_low.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ nano standard_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ gcc standard_copy.c -o standard_copy
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls
input.txt  low_level_copy  low_level_copy.c  output_low.txt  standard_copy  standard_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ./standard_copy input.txt output_standard.txt
File copied successfully using standard library functions.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cat output_standard.txt
This is Practical OSSP Session 9.
This file is used to test low-level file copying.
The program uses open, read, write, lseek, and close system calls.
It will also be compared with standard library functions.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp input.txt output_standard.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ time ./low_level_copy input.txt output_low.txt
File copied successfully using low-level system calls.

real    0m0.002s
user    0m0.002s
sys     0m0.000s
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ time ./standard_copy input.txt output_standard.txt
File copied successfully using standard library functions.

real    0m0.003s
user    0m0.001s
sys     0m0.002s
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ yes "OSSP Session 9 file copy performance testing" | head -c 100M > large_input.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls -lh large_input.txt
-rw-r--r-- 1 koumudi koumudi 100M Sep 24 09:41 large_input.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ time ./low_level_copy large_input.txt large_low.txt
File copied successfully using low-level system calls.

real    0m0.112s
user    0m0.004s
sys     0m0.108s
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ time ./standard_copy large_input.txt large_standard.txt
File copied successfully using standard library functions.

real    0m0.119s
user    0m0.012s
sys     0m0.105s
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp large_input.txt large_low.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$
```

**TASK2:** Develop a program that redirects standard input and Output using file descriptors and dup2(). Demonstrate how shells implement 1/0 redirection.

koumudi@Koumudi: ~/OSSP/

GNU nano 7.2 redir

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd;

    fd = open("redirect_output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    if (dup2(fd, STDOUT_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        return 1;
    }

    close(fd);

    printf("This output has been redirected to a file.\n");
    printf("Standard output is now connected to redirect_output.txt.\n");

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp large_input.txt large_low.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ nano redirect.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ gcc redirect.c -o redirect
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ./redirect./redirect
-bash: ./redirect./redirect: No such file or directory
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls
input.txt        large_low.txt       low_level_copy    output_low.txt       redirect    standard_copy
large_input.txt  large_standard.txt  low_level_copy.c  output_standard.txt  redirect.c  standard_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cat redirect_output.txt
cat: redirect_output.txt: No such file or directory
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ./redirect
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls
input.txt        large_low.txt       low_level_copy    output_low.txt       redirect    redirect_output.txt  standard_copy.c
large_input.txt  large_standard.txt  low_level_copy.c  output_standard.txt  redirect.c  standard_copy
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cat redirect_output.txt
This output has been redirected to a file.
Standard output is now connected to redirect_output.txt.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$
```

koumudi@Koumudi: ~/OSSP/

```
 GNU nano 7.2
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("input.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    if (dup2(fd, STDIN_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        return 1;
    }

    close(fd);

    read(STDIN_FILENO, buffer, sizeof(buffer) - 1);

    printf("Input read after redirecting stdin:\n");
    printf("%s\n", buffer);

    return 0;
}
```

koumudi@Koumudi: ~/OSSP/

```
This output has been redirected to a file.
Standard output is now connected to redirect_output.txt.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ nano input_redirect.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ gcc input_redirect.c -o input_redirect
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ./input_redirect
Input read after redirecting stdin:
This is Practical OSSP Session 9.
This file is used to test low-level file copying.
The program use
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ echo "Hello from shell redirection" > shell_output.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cat shell_output.txt
Hello from shell redirection
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls -lh
total 301M
-rw-r--r-- 1 koumudi koumudi  209 Sep 24 09:23 input.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:52 input_redirect
-rw-r--r-- 1 koumudi koumudi  518 Sep 24 09:50 input_redirect.c
-rw-r--r-- 1 koumudi koumudi 100M Sep 24 09:41 large_input.txt
-rw-r--r-- 1 koumudi koumudi 100M Sep 24 09:42 large_low.txt
-rw-r--r-- 1 koumudi koumudi 100M Sep 24 09:42 large_standard.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:27 low_level_copy
-rw-r--r-- 1 koumudi koumudi 1.3K Sep 24 09:25 low_level_copy.c
-rw-r--r-- 1 koumudi koumudi  209 Sep 24 09:31 output_low.txt
-rw-r--r-- 1 koumudi koumudi  209 Sep 24 09:31 output_standard.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:46 redirect
-rw-r--r-- 1 koumudi koumudi  533 Sep 24 09:46 redirect.c
-rw-r--r-- 1 koumudi koumudi  100 Sep 24 09:47 redirect_output.txt
-rw-r--r-- 1 koumudi koumudi   29 Sep 24 09:55 shell_output.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:30 standard_copy
-rw-r--r-- 1 koumudi koumudi 1.1K Sep 24 09:30 standard_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cat redirect_output.txt
This output has been redirected to a file.
Standard output is now connected to redirect_output.txt.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cat shell_output.txt
Hello from shell redirection
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ls -lh
total 301M
-rw-r--r-- 1 koumudi koumudi  209 Sep 24 09:23 input.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:52 input_redirect
-rw-r--r-- 1 koumudi koumudi  518 Sep 24 09:50 input_redirect.c
```

```
-rw-r--r-- 1 koumudi koumudi  209 Sep 24 09:23 input.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:52 input_redirect
-rw-r--r-- 1 koumudi koumudi  518 Sep 24 09:50 input_redirect.c
-rw-r--r-- 1 koumudi koumudi 100M Sep 24 09:41 large_input.txt
-rw-r--r-- 1 koumudi koumudi 100M Sep 24 09:42 large_low.txt
-rw-r--r-- 1 koumudi koumudi 100M Sep 24 09:42 large_standard.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:27 low_level_copy
-rw-r--r-- 1 koumudi koumudi 1.3K Sep 24 09:25 low_level_copy.c
-rw-r--r-- 1 koumudi koumudi  209 Sep 24 09:31 output_low.txt
-rw-r--r-- 1 koumudi koumudi  209 Sep 24 09:31 output_standard.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:46 redirect
-rw-r--r-- 1 koumudi koumudi  533 Sep 24 09:46 redirect.c
-rw-r--r-- 1 koumudi koumudi  100 Sep 24 09:47 redirect_output.txt
-rw-r--r-- 1 koumudi koumudi   29 Sep 24 09:55 shell_output.txt
-rwxr-xr-x 1 koumudi koumudi  16K Sep 24 09:30 standard_copy
-rw-r--r-- 1 koumudi koumudi 1.1K Sep 24 09:30 standard_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ grep lseek low_level_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$
```

```
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ grep lseek low_level_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ nano low_level_copy.c
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ gcc low_level_copy.c -o low_level_copy
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ ./low_level_copy large_input.txt large_low.txt
File copied successfully using low-level system calls.
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp large_input.txt large_low.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp large_input.txt large_standard.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ grep lseek low_level_copy.c
    /* Use lseek() to position the file at the beginning */
    if (lseek(source_fd, 0, SEEK_SET) == -1)
        perror("lseek");
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp large_input.txt large_low.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$ cmp large_input.txt large_standard.txt
koumudi@Koumudi:~/OSSP/PRACTICAL/Session_9$
```